

因果推断模型与机器学习方法

郑老师团队

时间: 2026

“因果推断模型与机器学习方法”——实践篇

倾向得分方法控制混杂偏倚的R实践

“论文与统计分析”公众号

“统计分析”公众号

倾向得分方法的医学应用案例

The NEW ENGLAND JOURNAL of MEDICINE

ORIGINAL ARTICLE

Effectiveness and Safety of Drug-Eluting Stents in Ontario

Jack V. Tu, M.D., Ph.D., James Bowen, B.Sc.Pharm., M.Sc., Maria Chiu, M.Sc.,
Dennis T. Ko, M.D., M.Sc., Peter C. Austin, Ph.D., Yaohua He, M.D., Ph.D.,
Robert Hopkins, M.A., Jean-Eric Tarride, Ph.D., Gord Blackhouse, M.B.A., M.Sc.,
Charles Lazzam, M.D., Eric A. Cohen, M.D., and Ron Goeree, M.A.

医学应用案例

研究总览

- **研究目的：**探讨药物洗脱架相对于裸金属支架防治心血管病的效果
- **数据来源：**加拿大心脏护理网络的真实世界数据
- **暴露：**是否安装药物洗脱支架（暴露组5106例，非暴露组8247例）
- **主要结局：**2年内是否再次心血管手术及其发生时间，是否心肌梗死、是否死亡及其发生时间
- **研究结论：**药物洗脱支架在降低心血管再狭窄高风险患者的再次心血管手术上具有显著效果，且不会显著增加死亡率或心肌梗死发生率。

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原始数据中暴露组、对照组的基线分布情况

- We compared the prevalences of these risk factors in the bare-metal–stent group and the drug-eluting–stent group before the propensity-score matching, using a **t-test** for means and **chi-square** tests for percentages.

Table 1. Baseline Characteristics of Patients, According to Stent Group, before Propensity-Score Matching.*

Characteristic	Bare-Metal Stent (N=8247)	Drug-Eluting Stent (N=5106)	P Value
Age (yr)	62.6±11.8	61.7±11.5	<0.001
Male sex (%)	74.4	70.0	<0.001
Income quintile (%)†			
1	18.8	18.7	0.89
2	20.3	19.9	
3	20.7	21.2	
4	21.5	21.2	
5	18.6	19.0	
Cardiac condition or procedure (%)			
Hypertension	37.1	36.4	0.45
Myocardial infarction			
Same day as index PCI	13.7	7.4	<0.001
1–7 days before index PCI	22.3	18.3	
8–365 days before index PCI	11.3	12.7	
None within 365 days before index PCI	52.7	61.5	
CCS angina classification‡			
0	7.5	6.4	<0.001
I	4.8	5.6	
II	13.8	15.6	
III	20.6	26.2	
IVA	28.4	25.2	
IVB	12.1	9.4	
IVC	12.1	9.4	
IVD	1.7	0.7	

可以看到：很多指标在两组之间的数值差异较大，并且差异具有显著性。

Congestive heart failure	5.0	5.4	0.28
Previous coronary-artery bypass surgery	7.7	9.5	<0.001
PCI >1 yr before index PCI	4.4	6.1	<0.001
Coexisting condition (%)			
Diabetes	24.5	38.0	<0.001
Peripheral vascular disease	5.7	5.8	0.96
Chronic obstructive pulmonary disease	5.3	4.0	<0.001
Cerebrovascular disease	3.6	5.9	<0.001
Primary cancer	1.1	0.9	0.52
Renal failure requiring dialysis	0.8	1.3	0.005
Index PCI			
Ad hoc PCI (%)	58.7	50.5	<0.001
Stent length (mm)	24.7±15.3	28.7±16.8	<0.001
Stent diameter (mm)	3.1±0.5	2.8±0.4	<0.001
No. of stents per patient	1.5±0.8	1.5±0.8	0.98
No. of vessels stented	1.1±0.4	1.1±0.4	0.17
ACC–AHA lesion type (%)			
A	13.2	6.4	<0.001
B1	29.6	25.9	
B2	36.0	37.8	
C	21.1	29.8	

倾向得分匹配方法的应用

1. 构建统计模型，计算倾向得分

- ✓ 首先单因素回归分析，筛选与三个主要结局指标相关的因素，共筛选得到21个潜在混杂因素
- ✓ 以是否安装药物洗脱支架为因变量，这21个因素为自变量，构建Logistic回归模型，计算所有个体的倾向得分。

Before performing propensity-score matching, we conducted a **univariate analysis** of all potentially available explanatory factors **for target-vessel revascularization, myocardial infarction, or death**. A total of **21 factors** were identified that were significantly associated with one or more of these three primary outcomes ($P < 0.05$).

倾向得分匹配方法的应用

2. 根据倾向得分，两组间匹配样本

- ✓ 以倾向得分为依据，通过匹配方法来凑出配对个体。
- ✓ 采用最邻近匹配法+卡钳值，配对比是1:1

A nearest-neighbor-matching algorithm with a “greedy” heuristic..... Matching occurring if the difference in the logits of the propensity scores was less than 0.2 times the standard deviation of the scores (the caliper width).

After propensity-score matching was completed, there were 3751 matched pairs of patients. (原始数据的两组人数分别是8247、5106)

倾向得分匹配方法的应用

3. 分析协变量均衡性以检验匹配效果

- ✓ 计算标准化均差 (SMD值)

We computed the standardized difference between the two groups for each type of distribution, with differences of **less than 0.1** taken to indicate good balance in the matched cohort.

- ✓ 计算组间协变量分布的假设检验P值

The distributions of categorical variables and continuous variables between the two groups in the matched cohort were compared with the use of **McNemar's test** and **paired t-tests**, respectively.

医学应用案例

倾向得分匹配后两组协变量分布展示:

Table 2. Baseline Characteristics of Patients, According to Stent Group, after Propensity-Score Matching.^a

Characteristic	Bare-Metal Stent (N=3751)	Drug-Eluting Stent (N=3751)
Age (yr)	62.3±11.7	62.3±11.5
Male sex (%)	70.7	71.2
Income quintile (%)†		
1	19.1	19.5
2	19.5	20.2
3	21.3	20.3
4	21.2	21.2
5	18.8	18.8
Cardiac condition or procedure (%)		
Hypertension	36.6	36.7
Myocardial infarction		
Same day as index PCI	9.1	9.2
1–7 days before index PCI	19.5	19.8
8–365 days before index PCI	12.7	11.9
None within 365 days before index PCI	58.7	59.2
CCS angina classification‡		
0	7.3	6.6
I	5.5	5.4
II	15.1	15.0
III	23.3	23.7
IVA	27.0	26.6
IVB	11.1	11.2
IVC	9.7	10.6
IVD	1.0	0.9
Congestive heart failure	5.0	5.3
Previous coronary-artery bypass surgery	9.0	8.5
PCI >1 yr before index PCI	5.5	5.2
Coexisting condition (%)		
Diabetes	32.6	32.6
Peripheral vascular disease	6.3	5.6
Chronic obstructive pulmonary disease	4.7	4.6
Cerebrovascular disease	4.9	5.2
Primary cancer	1.1	1.1
Renal failure requiring dialysis	1.2	1.1

Table 2. (Continued.)

Characteristic	Bare-Metal Stent (N=3751)	Drug-Eluting Stent (N=3751)
Index PCI		
Ad hoc PCI (%)	53.4	53.7
Stent length (mm)	26.3±16.8	26.6±15.2
Stent diameter (mm)	2.8±0.4	2.8±0.4
No. of stents per patient	1.5±0.8	1.5±0.8
No. of vessels stented	1.1±0.4	1.1±0.4
ACC-AHA lesion type (%)		
A	7.6	8.2
B1	29.1	28.9
B2	38.0	37.5
C	25.2	25.4

- 组间协变量分布的假设检验P值

There were no significant differences between the drug-eluting-stent group and the bare-metal-stent group in any of the 21 characteristics ($P > 0.20$ for all comparisons).

- 标准化均差法 (SMD)

The standardized difference between the two groups was less than 0.03 for all variables.

医学应用案例

倾向得分匹配方法的应用

4. 用匹配人群开展所需要的统计描述和分析

- 绘制Kaplan-Meier曲线

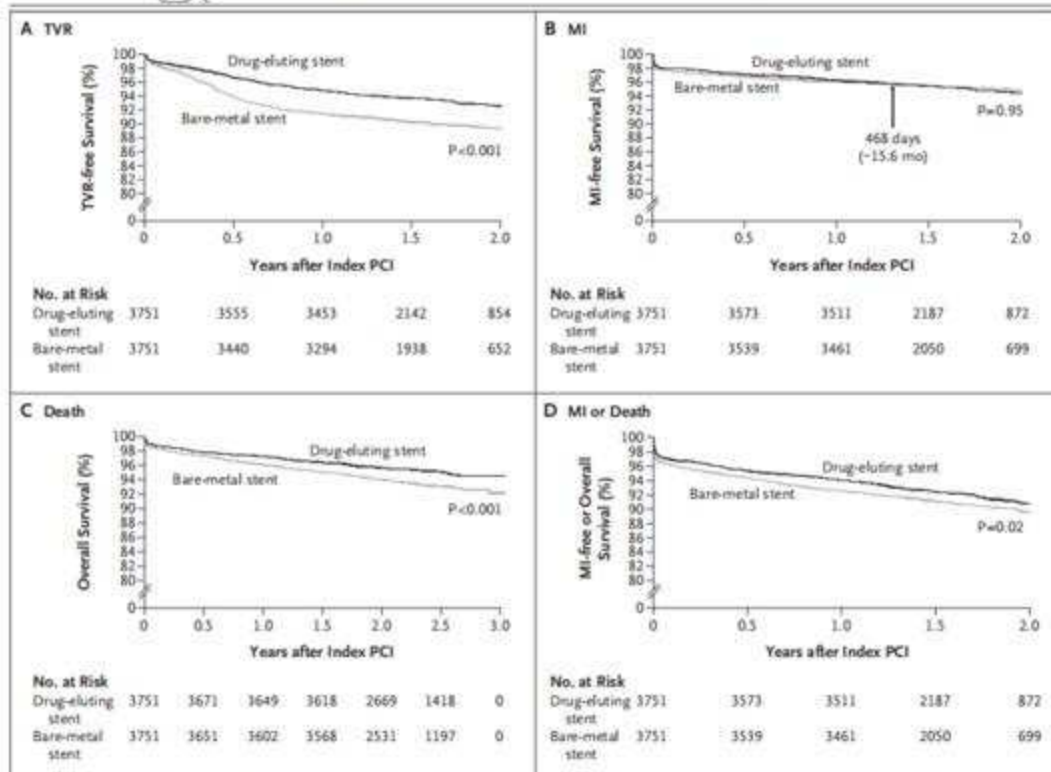


Figure 1. Event-free Survival in a Propensity-Score-Matched Cohort of Patients Who Received a Drug-Eluting Stent or a Bare-Metal Stent. Kaplan-Meier curves are shown for target-vessel revascularization (TVR) (Panel A), myocardial infarction (MI) (Panel B), overall survival (Panel C), and MI-free or overall survival (Panel D) after the index percutaneous coronary intervention (PCI).

- log-Rank检验

Table 3. Outcome Rates after Index PCI in the Propensity-Score-Matched Cohort, According to Subgroup.*

Outcome	At 6 Mo		At 1 Yr		At 2 Yr		At 3 Yr		P Value†
	Bare-Metal Stent	Drug-Eluting Stent	Bare-Metal Stent	Drug-Eluting Stent	Bare-Metal Stent	Drug-Eluting Stent	Bare-Metal Stent	Drug-Eluting Stent	
	percent								
Target-vessel revascularization	6.0	3.2	8.6	5.2	10.7	7.4			<0.001
Myocardial infarction	3.3	2.9	3.9	3.8	5.2	5.7			0.95
Death	2.7	2.1	4.0	2.7	6.1	4.3	7.8	5.5	<0.001
Myocardial infarction or death	5.7	4.8	7.5	6.1	10.5	9.3			0.02

* Outcome rates were derived from paired Kaplan-Meier curves. PCI denotes percutaneous coronary intervention.

† P values are for the comparisons of the paired Kaplan-Meier curves.

- 单因素Cox回归

Table 4. Rates of Target-Vessel Revascularization in the Propensity-Score-Matched Cohort, According to Subgroup.*

Diabetes	Vessel Diameter	Lesion Length	No. of Matched Pairs	Target-Vessel Revascularization		Hazard Ratio (95% CI)†	P Value	No. Needed to Treat
				Bare-Metal Stent	Drug-Eluting Stent			
				percent				
Yes	Small	Long	347	17.6	7.2	0.38 (0.24–0.60)	<0.001	10
Yes	Small	Short	253	13.0	4.7	0.34 (0.17–0.66)	0.002	12
Yes	Large	Long	295	10.5	6.1	0.55 (0.32–0.95)	0.03	23
Yes	Large	Short	291	7.6	6.2	0.78 (0.42–1.43)	0.42	71
No	Small	Long	782	12.3	8.6	0.66 (0.48–0.91)	0.01	27
No	Small	Short	502	8.0	6.8	0.82 (0.51–1.30)	0.40	83
No	Large	Long	638	7.5	5.6	0.74 (0.48–1.15)	0.18	53
No	Large	Short	505	5.9	5.3	0.87 (0.52–1.47)	0.61	167

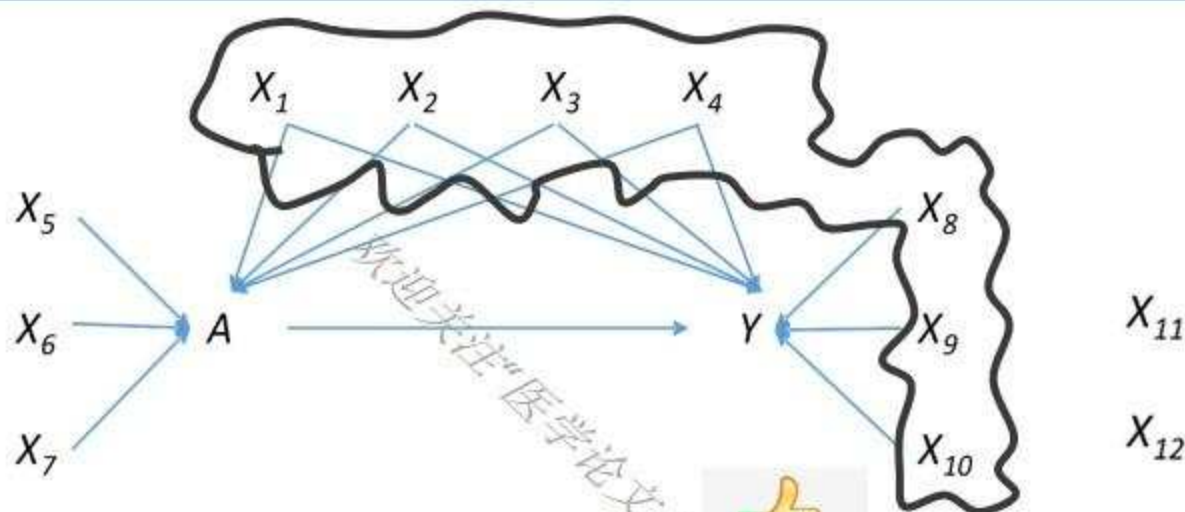
倾向得分方法的R语言实践

R语言实践

• 第一步：确定纳入倾向得分模型的协变量

协变量纳入考虑：

- ✓ 可以纳入所有协变量（协变量不多）
- ✓ 可以纳入仅和结局变量相关的协变量（单因素回归 $P < 0.05$ ，一般情况下**最推荐**）
- ✓ 可以纳入与暴露变量相关的变量（组间分布不均衡的协变量， $P < 0.05$ ）
- ✓ 可以同时纳入与暴露变量有关的变量、与结局有关的变量
- ✓ 纳入协变量并非多多益善，纳入不恰当容易估计不稳定
- ✓ 分类协变量在筛选之前要因子化factor



R语言实践

- 第一步：确定纳入倾向得分模型的协变量
- 第二步：构建统计模型，计算倾向得分
- 第三步：运用倾向得分，之后分析均衡性
 1. 第一种方式：倾向得分匹配
 2. 第二种运用方式：倾向得分分层
 3. 第三种运用方式：倾向得分校正
 4. 第四种运用方式：倾向得分加权（**最推荐方式**，R能做、SPSS目前还不能做）
- 第四步：开展统计分析



谢谢

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